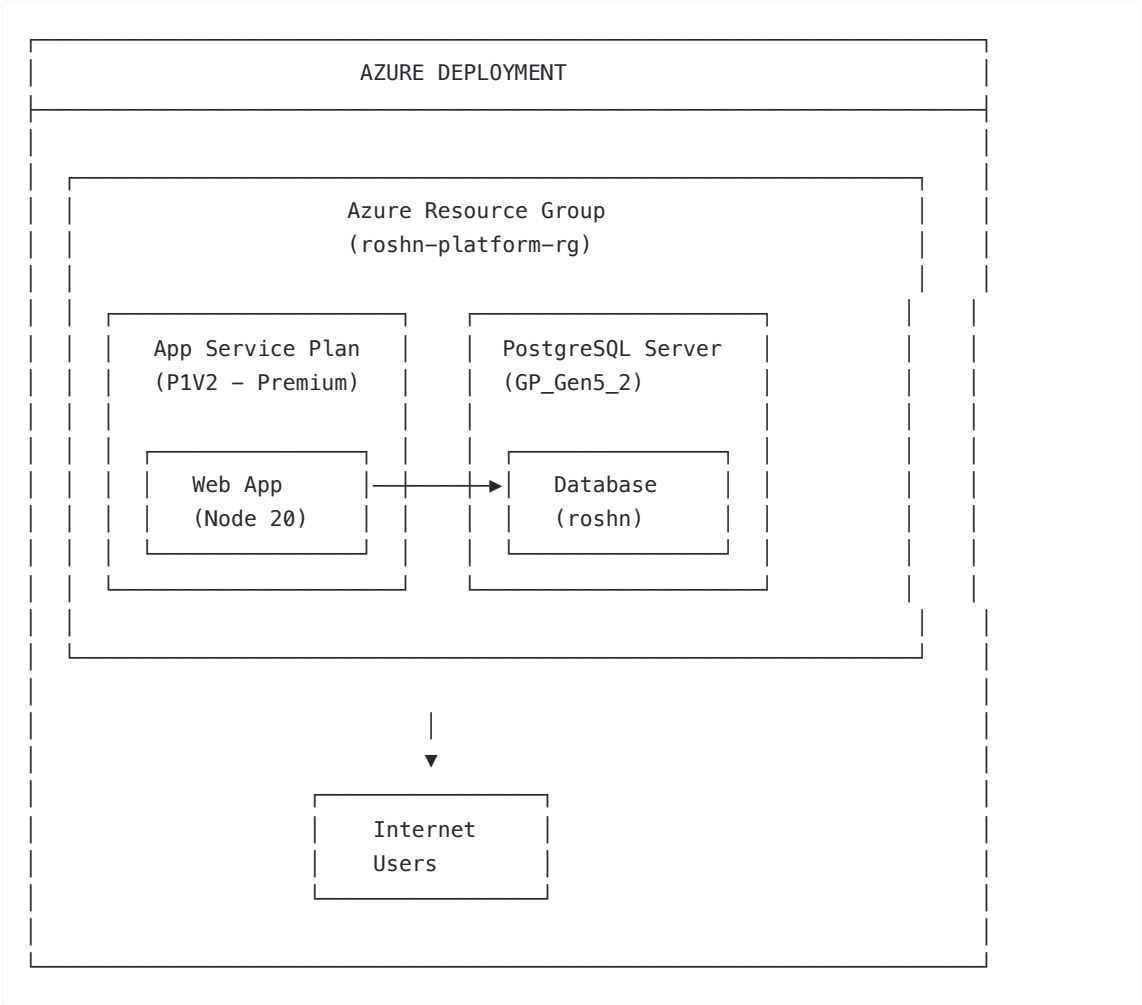


Azure Deployment Guide

Overview

This guide explains how to deploy the ROSHN Parametric Masterplan Modelling Platform to Microsoft Azure using Azure App Service and Azure Database for PostgreSQL.

Architecture



Prerequisites

Before deploying, ensure you have:

1. **Azure Account** with an active subscription
2. **Azure CLI** installed (version 2.40 or later)
3. **Node.js** 20.x or later
4. **Git** for version control

Install Azure CLI

macOS:

```
brew install azure-cli
```

Windows:

```
winget install Microsoft.AzureCLI
```

Linux:

```
curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
```

Deployment Options

Option 1: Automated Script Deployment

Use the provided deployment script for a complete automated setup:

```
# Clone the repository
git clone https://github.com/tajalagawani/roshn.git
cd roshn

# Make the script executable
chmod +x scripts/deploy-azure.sh

# Deploy to specific environment
./scripts/deploy-azure.sh dev      # Development environment
./scripts/deploy-azure.sh staging  # Staging environment
./scripts/deploy-azure.sh prod    # Production environment (default)
```

Environment Configurations

Environment	App Service SKU	Database SKU	Description
dev	B1 (Basic)	B_Gen5_1 (Basic)	Low-cost development
staging	S1 (Standard)	GP_Gen5_2	Testing environment
prod	P1V2 (Premium)	GP_Gen5_4	Full production

The script will:

1. Create a Resource Group
2. Create PostgreSQL Server and Database
3. Create App Service Plan and Web App
4. Configure environment variables
5. Deploy the application
6. Run database migrations

Option 2: Manual Deployment

Follow these steps for manual deployment:

Step 1: Login to Azure

```
az login
```

Step 2: Create Resource Group

```
az group create \  
  --name roshn-platform-rg \  
  --location uaenorth
```

Step 3: Create PostgreSQL Server

```
az postgres server create \  
  --resource-group roshn-platform-rg \  
  --name roshn-db-server \  
  --location uaenorth \  
  --admin-user roshnadmin \  
  --admin-password "YourSecurePassword123!" \  
  --sku-name GP_Gen5_2 \  
  --version 14
```

Step 4: Configure Firewall

```
az postgres server firewall-rule create \  
  --resource-group roshn-platform-rg \  
  --server roshn-db-server \  
  --name AllowAzureServices \  
  --start-ip-address 0.0.0.0 \  
  --end-ip-address 0.0.0.0
```

Step 5: Create Database

```
az postgres db create \  
  --resource-group roshn-platform-rg \  
  --server-name roshn-db-server \  
  --name roshn_platform
```

Step 6: Create App Service Plan

```
az appservice plan create \  
  --name roshn-platform-plan \  
  --resource-group roshn-platform-rg \  
  --sku P1V2 \  
  --is-linux
```

Step 7: Create Web App

```
az webapp create \  
  --resource-group roshn-platform-rg \  
  --plan roshn-platform-plan \  
  --name roshn-platform \  
  --runtime "NODE:20-lts"
```

Step 8: Configure Environment Variables

```
az webapp config appsettings set \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --settings \  
    DATABASE_URL="postgres://roshnadmin@roshn-db-server:PASSWORD@roshn-db-  
server.postgres.database.azure.com:5432/roshn_platform?sslmode=require" \  
    NEXTAUTH_SECRET="your-secret-key" \  
    NEXTAUTH_URL="https://roshn-platform.azurewebsites.net" \  
    NODE_ENV="production"
```

Step 9: Deploy Application

```
# Build the application  
npm ci  
npm run build  
  
# Create deployment package  
zip -r deploy.zip .next package.json package-lock.json public node_modules prisma  
  
# Deploy  
az webapp deployment source config-zip \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --src deploy.zip
```

Step 10: Run Migrations

```
npx prisma migrate deploy  
npx prisma db seed
```

Environment Variables

Variable	Description	Required
DATABASE_URL	PostgreSQL connection string	Yes
NEXTAUTH_SECRET	Secret for NextAuth.js encryption	Yes

NEXTAUTH_URL	Full URL of the application	Yes
NODE_ENV	Environment (production)	Yes
NEXT_TELEMETRY_DISABLED	Disable Next.js telemetry	No

Azure Resources Created

Resource	Name	SKU/Size	Purpose
Resource Group	roshn-platform-rg	-	Container for all resources
App Service Plan	roshn-platform-plan	P1V2	Hosting plan for web app
Web App	roshn-platform	Node 20 LTS	Application hosting
PostgreSQL Server	roshn-db-server	GP_Gen5_2	Database server
PostgreSQL Database	roshn_platform	-	Application database

Recommended Azure Regions

Region	Location	Code
UAE North	Dubai	uaenorth
UAE Central	Abu Dhabi	uaecentral
West Europe	Netherlands	westeurope
North Europe	Ireland	northeurope

For ROSHN projects, **UAE North** is recommended for optimal performance.

Scaling

Vertical Scaling (Scale Up)

Increase resources for the App Service Plan:

```
az appservice plan update \  
  --name roshn-platform-plan \  
  --resource-group roshn-platform-rg \  
  --sku P2V2
```

Available SKUs:

SKU	vCPU	Memory
B1	1	1.75 GB

B2	2	3.5 GB
P1V2	1	3.5 GB
P2V2	2	7 GB
P3V2	4	14 GB

Horizontal Scaling (Scale Out)

Add more instances:

```
az webapp update \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --set siteConfig.numberOfWorkers=3
```

Monitoring

View Logs

```
# Stream live logs  
az webapp log tail \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform  
  
# Download logs  
az webapp log download \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform
```

Enable Application Insights

```
# Create Application Insights  
az monitor app-insights component create \  
  --app roshn-platform-insights \  
  --location uaenorth \  
  --resource-group roshn-platform-rg  
  
# Get instrumentation key  
INSTRUMENTATION_KEY=$(az monitor app-insights component show \  
  --app roshn-platform-insights \  
  --resource-group roshn-platform-rg \  
  --query instrumentationKey -o tsv)  
  
# Add to app settings  
az webapp config appsettings set \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --
```

```
--settings  
APPLICATIONINSIGHTS_CONNECTION_STRING="InstrumentationKey=$INSTRUMENTATION_KEY"
```

Custom Domain & SSL

Add Custom Domain

```
az webapp config hostname add \  
  --webapp-name roshn-platform \  
  --resource-group roshn-platform-rg \  
  --hostname platform.roshn.sa
```

Enable SSL

```
# Create managed certificate  
az webapp config ssl create \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --hostname platform.roshn.sa  
  
# Bind certificate  
az webapp config ssl bind \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --certificate-thumbprint <thumbprint> \  
  --ssl-type SNI
```

Backup & Recovery

Configure Backup

```
az webapp config backup create \  
  --resource-group roshn-platform-rg \  
  --webapp-name roshn-platform \  
  --backup-name "daily-backup" \  
  --container-url "https://yourstorageaccount.blob.core.windows.net/backups?  
sv=..."
```

Database Backup

PostgreSQL backups are automatic. To create manual backup:

```
pg_dump -h roshn-db-server.postgres.database.azure.com \  
  -U roshnadmin@roshn-db-server \  
  -d roshn_platform > backup.sql
```

Troubleshooting

Application Not Starting

1. Check logs: `az webapp log tail --resource-group roshn-platform-rg --name roshn-platform`
2. Verify environment variables are set correctly
3. Ensure database connection is working

Database Connection Issues

1. Verify firewall rules allow Azure services
2. Check connection string format
3. Ensure SSL mode is set to `require`

Slow Performance

1. Check App Service Plan SKU (upgrade if needed)
 2. Enable Application Insights for diagnostics
 3. Review database query performance
-

Cost Optimization

Development Environment

Use lower SKUs for non-production:

```
# Development App Service Plan
az appservice plan create \
  --name roshn-dev-plan \
  --resource-group roshn-dev-rg \
  --sku B1 \
  --is-linux

# Development Database
az postgres server create \
  --sku-name B_Gen5_1 \
  ...
```

Auto-Shutdown

Configure auto-shutdown for development resources to save costs.

Security Best Practices

1. Enable HTTPS Only

```
az webapp update \
  --resource-group roshn-platform-rg \
  --name roshn-platform \
  --https-only true
```


2. Enable Managed Identity

```
az webapp identity assign \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform
```

3. Configure IP Restrictions

```
az webapp config access-restriction add \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --rule-name "OfficeOnly" \  
  --priority 100 \  
  --ip-address "YOUR.OFFICE.IP.ADDRESS/32"
```

4. Enable TLS 1.2 Minimum

```
az webapp config set \  
  --resource-group roshn-platform-rg \  
  --name roshn-platform \  
  --min-tls-version 1.2
```